

JOY KUMAR GHOSH

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Profile Summary

CSE Lab Instructor (Part-Time) at North South University (NSU), where I also completed my B.Sc. in Computer Science and Engineering. First author of a published article in **APL Quantum (AIP Publishing)** on a hybrid quantum key distribution protocol, achieving a 43.67% improvement in key-generation rate over baseline. Hands-on experience in Qiskit-based circuit simulation on IBM Quantum Runtime hardware. Seeking graduate study opportunities in quantum cryptography, quantum key distribution, and quantum computing.

Research Interests

- Quantum Computing
- Quantum Cryptography
- Quantum Key Distribution
- Quantum Communication

Education

North South University (NSU) B.Sc. in Computer Science and Engineering Distinction: Magna Cum Laude CGPA: 3.77/4.00 ($\approx$ 94% marks)	Jan 2022 – Sep 2025 Dhaka, Bangladesh
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Research Experience

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| Undergraduate Research – North South University (NSU)
<ul style="list-style-type: none">Designed an incrementally-biased Ry-based selection schedule for a hybrid GHZ-BB84 QKD architecture, achieving a 43.67% improvement in key-generation rate over a fixed 50/50 baseline, under the supervision of Dr. Mahdy Rahman Chowdhury.Result published as first-author manuscript in APL Quantum (AIP Publishing). | Spring 2025 |
| Independent Research – Mahdy Research Academy
<ul style="list-style-type: none">Completed the thesis phase of an intensive private research program in Quantum Computing & Information, as part of a 2-member research team under the supervision of Dr. Mahdy Rahman Chowdhury.Authored a complete manuscript draft (submitted as “Accelerating Noisy Quantum Circuit Simulation via Unitary Precomputation”) and a research poster summarizing the findings. | August 2024 – Dec 2024 |

Peer-Reviewed Journal Publications

- Joy Kumar Ghosh, Arman Sykot, Most Atyea Sanjeeda Ema, Mahdy Rahman Chowdhury. “Enhanced Selection Schedule for Combined Quantum Key Distribution Protocol.” APL Quantum, 3(3), 2026.**
DOI: [10.1063/5.0318676](https://doi.org/10.1063/5.0318676)

Teaching Experience

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| CSE Lab Instructor – North South University (NSU)
<ul style="list-style-type: none">Teaching 6 lab sections across 2 courses, $\sim$240 student's total:<ul style="list-style-type: none">CSE225L (Data Structures & Algorithms Lab) – 4 SectionsCSE311L (Database Management System Lab) – 2 SectionDelivering hands-on instruction in MySQL/MariaDB (XAMPP, phpMyAdmin) and core data structures in C++ (arrays, linked lists, stacks, queues, trees, graphs).Authored CSE311L lab manuals featuring problem-first scenarios. | June 2026 – Present
Dhaka, Bangladesh |
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Industry Experience

Oracle APEX Web Application Developer – Intern
MySoft Limited

(4 Months) Aug 2025 – Nov 2025
Dhaka, Bangladesh

- Developed Outpatient Appointment System using APEX, and PL/SQL. [[Live Demo](#)]
- Participated in SRS Meetings with Client.
- Part of Undergraduate Program under the supervision of **Dr. Abu Sayed Md. Latiful Hoque**.

Selected Projects

1. **Undergraduate CSE445 (Machine Learning) Course Project:** A thorough examination of Dhaka City's weather data in Bangladesh, along with predictions of temperature, humidity, solar energy, solar radiation, and conditions through machine learning algorithms. Summer 2025. [[Report](#)]
2. **Undergraduate CSE465 (Pattern Recognition and Neural Network) Course Project:** Tomato Leaf Disease Detection using CNN and Lime as XAI. Spring 2025. [[Report](#)] [[Repository](#)]
3. **Undergraduate CSE425 (Concepts of Programming Language) Course Project:** Bangla Programming Language, Compiler Design with ANTLR. Summer 2024. [[Report](#)] [[Repository](#)]
4. **Undergraduate CSE311 (Database Systems) Course Project:** Nationwide Centralized Hospitals & Patients Management System, with MySQL, and PHP. Summer 2023. [[Report](#)] [[Repository](#)]

Technical Skills

- **Quantum Tools:** Qiskit, PennyLane, IBM Quantum Runtime
- **Programming Language:** Python, C/C++, Java.
- **Machine Learning Tools:** PyTorch, Scikit-learn, Numpy, Pandas, LIME.
- **Version Control:** Git/GitHub.
- **Writing Tools:** LaTeX, Microsoft/Google Office Suite
- **Language:** English (Professional), Bangla (Native).

Selected Certifications

1. **Mahdy Research Academy:** Quantum Computing & Information. [[Certificate](#)]
2. **University of Colorado:** Applied Cryptography, Specialization. [[Certificate](#)]
3. **Duke University:** Explainable AI (XAI), Specialization [[Certificate](#)]
4. **KAIST:** Introduction to Quantum Information. [[Certificate](#)]

Honors & Awards

1. **Magna Cum Laude**, awarded by North South University. **Sep 2025**
2. **Merit-Based Scholarship for Academic Excellence**, North South University — progressively increased: 25% (Spring 2023) → 50% (Spring 2024) → 75% (Summer 2024) → 100% (Summer 2025, final semester) **Spring 2023 – Summer 2025**

References

Dr. Mahdy Rahman Chowdhury
Professor, ECE Department,
North South University, Dhaka.
Email: mahdy.chowdhury@northsouth.edu

Dr. Abu Sayed Md. Latiful Hoque
Professor, Dept. of CSE, BUET;
Visiting Professor, Dept. of ECE, North South University
Email: asmlatifulhoque@cse.buet.ac.bd